

Opening - Greeting, Self-Introduction and Roadmap

Speaking Script

- Good morning or good afternoon to the panel. Thank you for the opportunity to present my final year project.
- My project is titled Coral Health AI, an accurate, explainable, and deployable deep learning framework for coral reef health assessment.
- The aim is to classify coral images into Healthy, Bleached, and Dead, explain each prediction visually using Grad-CAM, and evaluate the model fairly across all classes.
- I will walk through the problem, objectives, and scope, then the literature, the methodology, the dataset, the model and its optimisation, the results and Grad-CAM explanations, the ablation study, and finally the conclusion and future work.

Suggested Timing (about 15 minutes)

- Introduction, problem, objectives, and scope: about 3 minutes (Slides 1 to 3).
- Literature review and research gap: about 2 minutes (Slides 4 to 5).
- Methodology, dataset, and model development: about 4 minutes (Slides 6 to 8).
- Evaluation, Grad-CAM, and results: about 4 minutes (Slides 9 to 13).
- Ablation study, conclusion, and future work: about 2 minutes (Slides 14 to 15).
- Keep the remaining time for the panel question and answer session.

Expected Panel Questions With Answers

- **Q1:** In one sentence, what is your project about?
- **Answer:** It is a deep learning system that classifies coral images as Healthy, Bleached, or Dead, explains the decision visually using Grad-CAM, and reaches 98.11 percent accuracy on unseen test images.
- **Justification:** A clear one-line summary shows command of the whole project.

- **Q2:** What motivated you to choose this topic?
- **Answer:** Coral reefs are declining quickly, and manual reef monitoring is slow and depends on experts, so an automated and trustworthy assessment tool can support faster conservation decisions.
- **Justification:** This links the project to a real environmental need.

Slide 1 - Problem Statement

Speaking Script

- Coral reefs are among the most valuable marine ecosystems, but they are declining rapidly due to climate change and ocean warming, so assessing their health quickly and reliably has become urgent.
- My project addresses three engineering problems in this area.
- The first is that manual assessment by experts is slow, subjective, and cannot scale to the thousands of reef images that monitoring now produces.
- The second is that even when a CNN predicts correctly, it behaves like a black box, so users cannot see why a decision was made and therefore cannot fully trust or validate it.
- The third is that relying on accuracy alone can be misleading, because a model can score high overall while quietly failing on a smaller class such as Dead coral.
- So this project proposes one framework that is accurate, explainable, and properly evaluated, to solve all three problems together.

Expected Panel Questions With Answers

- **Q1:** Why is manual coral assessment a problem?
- **Answer:** Because it takes time, needs trained experts, and cannot easily handle many images.
- **Justification:** This supports the need for an automated system.

- **Q2:** Why is model interpretability important?
- **Answer:** Because users need to understand which coral region influenced the prediction.
- **Justification:** This helps build trust in the AI output.

- **Q3:** Why is accuracy alone not enough?
- **Answer:** Because a model can have high accuracy but still perform weakly on a smaller class.
- **Justification:** This is important because the Dead class has fewer images.

- **Q4:** Who will use this system, and why does explainability matter for them?
- **Answer:** The users are marine biologists, reef managers, and trained divers who must act on the result, such as flagging a reef for closer inspection. They are more

likely to trust and verify a decision when the system also shows which coral region caused it.

- **Justification:** This connects the model-trust problem to the real decision makers.

Slide 2 - Project Objectives

Speaking Script

- From those problems I set three core objectives, with deployment as a fourth applied objective in the report.
- The first objective is to develop a deep learning model based on EfficientNetB0, enhanced with Stochastic Weight Averaging and a five-seed ensemble, for classifying coral health.
- The second objective is to apply Grad-CAM as a visual explanation method, so each prediction can be interpreted and validated rather than trusted blindly.
- The third objective is to evaluate the model properly using accuracy, precision, recall, and F1-score on a held-out test set, so performance is measured fairly across every class.
- The fourth objective, covered in my report, is to deploy the model as a web application with real-time inference and Grad-CAM display.
- Together these objectives make the model accurate, explainable, reliable, and usable.

Expected Panel Questions With Answers

- **Q1:** Why did you choose EfficientNetB0?
- **Answer:** Because it gives a good balance between accuracy and computational efficiency.
- **Justification:** It is suitable for image classification without being too heavy.

- **Q2:** Why is Grad-CAM part of the objective?
- **Answer:** Because it shows where the model focuses in the image.
- **Justification:** This makes the result easier to explain to users and panels.

- **Q3:** Why use precision, recall, and F1-score?
- **Answer:** Because they show class-level performance, not only overall correctness.
- **Justification:** This gives a fairer evaluation for all coral classes.

- **Q4:** How will you prove that each objective was actually achieved?
- **Answer:** Objective one is measured by accuracy, precision, recall, and F1-score on the test set; objective two by Grad-CAM heatmaps that focus on coral regions; and objective three by reporting all metrics together with the confusion matrix and an ablation study.

- **Justification:** This makes the objectives measurable instead of only descriptive.

Slide 3 - Project Scope

Speaking Script

- The scope defines exactly what the project covers and what it does not.
- The data source is the BHD Coral Dataset from Kaggle, and the task is to classify each image into one of three health conditions: Healthy, Bleached, or Dead.
- Technically the scope runs end to end: image pre-processing, augmentation, model training, evaluation, Grad-CAM explanation, and finally deployment.
- The deployment is a locally hosted Flask web application with drag-and-drop upload, real-time classification, a confidence score, and a Grad-CAM overlay.
- All training was performed on a local RTX 3070 GPU to keep the computational conditions consistent.
- Importantly, the project is limited to image-based, whole-image classification. It does not measure water quality or temperature, does not use live video or sensor data, and does not identify coral species or diagnose disease beyond the three classes.

Expected Panel Questions With Answers

- **Q1:** Why did you limit the project to three classes?
- **Answer:** Because the dataset and project goal focus on the three main coral health conditions.
- **Justification:** This keeps the classification task clear and measurable.

- **Q2:** What is outside your project scope?
- **Answer:** It does not measure water quality, temperature, or real-time underwater sensor data.
- **Justification:** The project is focused on image classification.

- **Q3:** Why include deployment in the scope?
- **Answer:** Because deployment shows the model can be used in a real interface, not only in experiments.
- **Justification:** This improves the practical value of the project.

- **Q4:** A single photo can show both healthy and bleached coral. How does image-level classification handle mixed scenes?
- **Answer:** The current scope predicts one dominant health condition per image, so strongly mixed scenes are a known limitation. Patch-based classification or segmentation is identified as future work to handle them.

- **Justification:** This shows I understand the boundary of the chosen method.

Slide 4 - Literature Review & Research Gap

Speaking Script

- My literature review supports each major design choice in the project.
- EfficientNetB0 is supported as the backbone because studies show it is highly accurate yet efficient, reaching strong results with far fewer parameters than ResNet or VGG.
- Ensemble learning is supported because combining several models reduces the random errors of any single model.
- Multi-scale test-time augmentation is supported because evaluating an image at different scales and orientations gives more reliable predictions.
- And Grad-CAM is supported as a visual explainability method that works naturally with convolutional networks.
- The research gap is that previous studies usually applied these techniques separately. To my knowledge, no prior coral study combines EfficientNetB0, five-seed SWA, multi-scale TTA, and Grad-CAM into one framework for three-class coral health assessment, and that combination is my contribution.

Expected Panel Questions With Answers

- **Q1:** What is the research gap?
- **Answer:** No previous work in the reviewed studies combines EfficientNetB0, five-seed SWA, multi-scale TTA, and Grad-CAM into one coral framework.
- **Justification:** This defines the contribution of the project.

- **Q2:** Why is combining methods useful?
- **Answer:** Each method improves a different part: accuracy, stability, inference reliability, and explainability.
- **Justification:** The combination creates a stronger overall system.

- **Q3:** How is your project different from existing studies?
- **Answer:** It focuses on three-class coral health classification with both performance evaluation and visual explanation.
- **Justification:** This makes the work more complete and project-specific.

- **Q4:** Combining four techniques adds complexity. How do you justify that the extra cost is worth it?

- **Answer:** Each technique targets a different and evidenced weakness, and the ablation study shows the full pipeline cuts test errors from 24 to 3, about 87.5 percent fewer, while inference is still light enough for a single GPU.
- **Justification:** This shows the combination is driven by measured benefit, not only novelty.

Slide 5 - Literature Review Summary

Speaking Script

- This table summarises the most relevant studies and links each one directly to my methodology.
- Works by Zhou and by Shao in 2024 show that EfficientNet-based models outperform heavier architectures on underwater and medical image tasks, which justifies my backbone choice.
- Other studies demonstrate that ensemble learning improves robustness over a single model, supporting my five-seed SWA strategy.
- Kandel and Castelli in 2021 show that test-time augmentation with average voting consistently improves CNN performance, which motivates my multi-scale TTA.
- And studies by Ghaffar and Choi, and by Korkmaz, confirm Grad-CAM as a reliable explainability method for CNNs.
- Most of these references are recent, within the last five years, and each maps to a specific technical decision, so my design is evidence-based rather than arbitrary.

Expected Panel Questions With Answers

- **Q1:** Why are these references relevant?
- **Answer:** They support the exact techniques used in my framework: EfficientNet, ensemble, TTA, and Grad-CAM.
- **Justification:** This links literature directly to methodology.

- **Q2:** Why include studies from other domains?
- **Answer:** Because methods like EfficientNet and Grad-CAM are general image AI methods, not limited to coral only.
- **Justification:** They still support the technical choice.

- **Q3:** What did the literature help you decide?
- **Answer:** It helped justify the architecture, optimisation method, and explainability method.
- **Justification:** This shows the design is evidence-based.

- **Q4:** How many references support your work, and how recent are they?
- **Answer:** The review is built on high-quality references, mostly published between 2021 and 2025, such as Ghaffar and Choi 2025, Shao 2024, Zhou 2024, Korkmaz

2025, Long 2025, Alghatani 2025, Kandel and Castelli 2021, and Oza 2024, each mapped to a specific design choice.

- **Justification:** Recent and relevant references show the work is current and evidence based.

Slide 6 - Overall Methodology

Speaking Script

- My methodology follows a ten-stage pipeline grouped into three phases.
- The first phase is data preparation. I acquire the BHD dataset, pre-process every image by converting it to RGB and resizing to 224 by 224, then split the data using stratified sampling into 80 percent training, 10 percent validation, and 10 percent test.
- A key point here is that the split is done before any augmentation, and augmentation is applied only to the training set, which prevents data leakage into validation or test.
- The second phase is model development. I train EfficientNetB0 with transfer learning, apply Stochastic Weight Averaging across five seeds, and tune the hyperparameters against validation accuracy, loss, and F1-score.
- The third phase is analysis and output. I evaluate the final ensemble with multi-scale TTA, generate Grad-CAM heatmaps to explain the predictions, and deploy everything through a Flask web application.
- So the workflow takes a raw underwater image all the way to an explained, deployed prediction.

Expected Panel Questions With Answers

- **Q1:** Why divide methodology into phases?
- **Answer:** It makes the workflow easier to understand from data to model to deployment.
- **Justification:** This supports clear project planning.

- **Q2:** Which phase is most important?
- **Answer:** All are important, but data preparation strongly affects model quality.
- **Justification:** Bad data preparation can reduce model performance.

- **Q3:** Why include Grad-CAM after evaluation?
- **Answer:** Because after knowing the model performs well, we also need to understand why it predicts that way.
- **Justification:** This supports trust and explainability.

- **Q4:** How do you make sure there is no data leakage between the training, validation, and test sets?

- **Answer:** The dataset is split by class using stratified sampling before any processing, and augmentation, oversampling, and class weighting are applied only to the training set, so the 159 test images are never seen during training or tuning.
- **Justification:** This rules out an inflated accuracy caused by leakage.

Slide 7 - Dataset & Data Preparation

Speaking Script

- The dataset contains 1,582 RGB coral images captured underwater, so they vary in resolution, lighting, and colour.
- It has three classes: 712 Healthy, 720 Bleached, but only 150 Dead images.
- This is the central challenge of the dataset: it is imbalanced, because the Dead class is under-represented at under 10 percent of the data.
- All images are resized to 224 by 224 to match EfficientNetB0's input, and after the stratified split the Dead class has only about 120 training images.
- To handle the imbalance I use three strategies together: hard-example oversampling, where the hardest Dead samples are duplicated up to 30 times; class weighting, where the Dead class receives an extra 1.3 times weight; and augmentation such as rotation, horizontal flip, and zoom.
- These are applied only to the training set, so the validation and test sets stay clean and the evaluation remains fair.

Expected Panel Questions With Answers

- **Q1:** Why resize images to 224 by 224?
- **Answer:** Because EfficientNetB0 commonly uses 224 by 224 input images.
- **Justification:** This keeps training consistent and efficient.

- **Q2:** Why is class imbalance a problem?
- **Answer:** The model may focus more on classes with many images and learn less from the smaller class.
- **Justification:** This can hurt Dead class prediction.

- **Q3:** Why use augmentation?
- **Answer:** It creates varied training examples from existing images.
- **Justification:** This helps the model generalise better to new images.

- **Q4:** The Dead class has only 150 images and just 15 in the test set. How reliable is the Dead recall measured on 15 samples?
- **Answer:** I treat the Dead result carefully, because with only 15 test images one error already changes recall by about 6.7 percent, so the value carries wide uncertainty. This is why I also report macro F1 and the confusion matrix, use oversampling and class weighting, and list collecting more Dead images as future work.

- **Justification:** This shows honesty about the limits of the minority class.
- **Q5:** Why did you set the Dead class weight to 1.3 instead of a much larger value?
- **Answer:** Because the Dead class is already heavily oversampled, a mild weight of 1.3 is enough to focus the model without making it over-predict Dead and become unstable. The value was tuned on the validation set.
- **Justification:** This explains the balancing choice instead of using an arbitrary number.

Slide 8 - Model Development & Optimisation

Speaking Script

- The model uses EfficientNetB0 as the feature extractor, initialised with ImageNet pretrained weights through transfer learning, so it begins with useful visual knowledge instead of learning from scratch.
- I replace the original classifier with a custom head: a global average pooling layer, a dropout layer at 0.4 to reduce overfitting, and a final dense layer with three outputs and a softmax for the three classes.
- Only the last 100 layers of the backbone are fine-tuned, so the model adapts to coral features without losing its general low-level knowledge.
- Two optimisation techniques are central. Stochastic Weight Averaging averages the model's weights over the final epochs, 26 to 30, producing a more stable model that sits in a flatter region of the loss landscape, which generalises better.
- I train five such models with different seeds and average their predictions, which cancels the randomness of any single run.
- Then at inference, multi-scale TTA averages four views per image across the five models for added robustness.

Expected Panel Questions With Answers

- **Q1:** What does SWA mean in simple terms?
- **Answer:** SWA averages good model weights from later epochs to make one more stable model.
- **Justification:** This reduces dependence on only one final epoch.

- **Q2:** Why use five seeds?
- **Answer:** Each seed trains the model from a slightly different starting point.
- **Justification:** Combining them reduces random mistakes.

- **Q3:** Why use TTA during inference?
- **Answer:** TTA runs each image at two sizes, 224 by 224 and 256 by 256 with the 256 image centre-cropped back to 224, each evaluated original and horizontally flipped. That gives four views per image, and the four views across the five ensemble models are averaged, so 20 predictions are combined into the final result.
- **Justification:** Averaging the views makes the prediction more stable.

- **Q4:** Why did you choose EfficientNetB0 over ResNet50, a larger EfficientNet, or a Vision Transformer?
- **Answer:** EfficientNetB0 gives the best accuracy for its small size, which suits a dataset of only about 1,500 images. Larger networks and Vision Transformers usually need much more data and tend to overfit small datasets, and B0 is also lighter to deploy.
- **Justification:** The architecture is matched to the data size and the deployment goal.
- **Q5:** Can you clearly explain the difference between SWA, the five-seed ensemble, and TTA?
- **Answer:** SWA averages the weights of one model over its last epochs, 26 to 30, into a single stable model. The five-seed ensemble trains five such models from different random seeds and averages their output probabilities. TTA works at prediction time by running each image at two sizes, 224 and 256 with the 256 centre-cropped to 224, each original and horizontally flipped, and averaging the four views across the five models.
- **Justification:** This shows I understand each technique precisely, not just by name.

Slide 9 - Model Evaluation

Speaking Script

- The model is evaluated on 159 unseen test images that were never used in training or tuning, so this is a fair test of how well it generalises.
- For each image the ensemble outputs probabilities for Healthy, Bleached, and Dead.
- I deliberately go beyond accuracy. I report precision, recall, F1-score, and a confusion matrix, and I focus on the macro-averaged F1.
- Macro averaging is important because it treats every class equally, so weak performance on the small Dead class cannot be hidden behind the larger Healthy and Bleached classes.
- I also report the weighted average to reflect overall performance across the realistic class distribution.
- Together these metrics give both an overall picture and a class-by-class picture, which is essential for an imbalanced problem like this one.

Expected Panel Questions With Answers

- **Q1:** Why use a held-out test set?
- **Answer:** Because it checks how the model performs on images it has never seen.
- **Justification:** This measures real generalisation.

- **Q2:** What does macro F1-score mean?
- **Answer:** It averages F1-score across all classes equally.
- **Justification:** This is useful when classes are imbalanced.

- **Q3:** Why use a confusion matrix?
- **Answer:** It shows exactly which classes are predicted correctly and which are confused.
- **Justification:** This helps explain the model errors clearly.

- **Q4:** Why do you report macro-averaged metrics instead of weighted or overall averages?
- **Answer:** Macro averaging treats every class equally, so weak performance on the small Dead class cannot be hidden by the larger Healthy and Bleached classes. A weighted average would be dominated by the bigger classes.
- **Justification:** Macro averaging is the fair choice for an imbalanced dataset.

Slide 10 - Grad-CAM Explainability

Speaking Script

- Grad-CAM is how I make the model explainable, which satisfies the second objective.
- It works by taking the gradients at the final convolutional layer, the top_conv layer of EfficientNetB0, and using them to find which regions most influenced the prediction.
- The result is a heatmap overlaid on the coral image, where red means the model focused strongly there and blue means low focus.
- Biologically I expect specific patterns: for Healthy coral the focus should be on coral texture and vibrant pigmentation; for Bleached coral on the pale, whitened surface; and for Dead coral on the exposed skeletal structure.
- This matters because a model can sometimes be right for the wrong reasons, so Grad-CAM lets me confirm the decision is based on real coral features rather than the background, which builds trust in the system.

Expected Panel Questions With Answers

- **Q1:** What does Grad-CAM actually show?
- **Answer:** It shows which image regions influenced the model decision most.
- **Justification:** This makes the AI decision easier to understand.

- **Q2:** How do you know the model is looking at the right area?
- **Answer:** The heatmap should appear on coral structures, not random background.
- **Justification:** This supports biological meaning.

- **Q3:** Why is explainability important for coral assessment?
- **Answer:** Because users need to trust that the model looks at coral health features.
- **Justification:** This improves confidence in the system.

- **Q4:** Grad-CAM can sometimes highlight regions that are not truly the cause. How do you know your explanation is trustworthy?
- **Answer:** Grad-CAM is a visual aid, not absolute proof, so I support it by checking that the heatmaps consistently land on coral structures rather than background, and by noting that the more focused model is also more accurate and more confident. Stronger faithfulness tests are listed as future work.
- **Justification:** This shows I understand the limits of explainable AI.

Slide 11 - Final Classification Results

Speaking Script

- These are the final results of the five-seed SWA ensemble on the 159-image test set.
- The model achieved 98.11 percent accuracy, correctly classifying 156 of 159 images, with macro precision of 98.63, recall of 96.85, and a macro F1 of 97.69 percent.
- Looking at the confusion matrix, the Healthy class achieved perfect recall, meaning every Healthy sample was found, and the Dead class achieved perfect precision, meaning the model never raised a false Dead alarm.
- Only three errors occurred: two Bleached corals were predicted as Healthy, and one Dead coral was predicted as Bleached.
- These errors are not random; they sit on the boundaries between visually similar health states, and importantly each one was made at low confidence, below 55 percent, so the model effectively flagged its own uncertainty.
- I emphasise macro F1 as the headline metric because it balances precision and recall fairly across all three classes, including the minority Dead class.

Expected Panel Questions With Answers

- **Q1:** What does 98.11 percent accuracy mean?
- **Answer:** It means 156 out of 159 unseen test images were classified correctly.
- **Justification:** This shows strong overall performance.

- **Q2:** Why did the model make three mistakes?
- **Answer:** Some coral images have similar visual patterns near class boundaries.
- **Justification:** This makes a few misclassifications expected.

- **Q3:** Which result is most important?
- **Answer:** Macro F1-score is very important because it balances precision and recall across all classes.
- **Justification:** This is fairer for imbalanced data.

- **Q4:** Two Bleached corals were predicted as Healthy. In conservation terms, is that not the most costly type of error?
- **Answer:** Yes, predicting a Bleached reef as Healthy is the most serious error, because a stressed reef could be missed. To reduce this risk, the system reports

per-class recall, shows the Grad-CAM heatmap and confidence so a human can verify, and in deployment can send low-confidence cases for expert review.

- **Justification:** This links a single confusion-matrix cell to real-world impact and a solution.
- **Q5:** 98 percent on a small test set looks very high. How do you defend against overfitting?
- **Answer:** The split is stratified and leakage free, and I use transfer learning, dropout, SWA, and ensembling to reduce variance, with the ablation study showing consistent gains across configurations. I still state the small Dead test set as a limitation.
- **Justification:** This answers the common concern that the result is too good to be true.

Slide 12 - Grad-CAM Explainability Results

Speaking Script

- This slide compares the Grad-CAM explanations of the baseline model against my proposed ensemble.
- The baseline heatmaps are diffuse and unreliable. In one striking case it classified a Healthy coral correctly, but Grad-CAM showed it was actually focusing on a clownfish in the foreground rather than the coral, so it was right for the wrong reason.
- Its confidence was also low, between roughly 45 and 75 percent.
- The proposed ensemble is clearly better. Its heatmaps concentrate on biologically meaningful regions: pigmented tissue for Healthy, the pale surface for Bleached, and exposed skeleton for Dead.
- Its confidence is also much higher, between about 97 and 99 percent on correct predictions.
- This is strong evidence that the ensemble is not only more accurate but more trustworthy, because it looks at the right features, which directly demonstrates that my second objective, explainability, has been achieved.

Expected Panel Questions With Answers

- **Q1:** Why are focused heatmaps better?
- **Answer:** Focused heatmaps show the model is looking at specific coral features.
- **Justification:** This is better than attention spread over background.

- **Q2:** How does this support objective two?
- **Answer:** Objective two is explainability using Grad-CAM, and this slide shows clearer visual explanations.
- **Justification:** It proves the objective was achieved.

- **Q3:** Does Grad-CAM improve accuracy?
- **Answer:** Grad-CAM itself explains the decision, while the ensemble improves prediction stability.
- **Justification:** They support different parts of the project.

- **Q4:** The confidence jumps are large, for example Bleached from 45.5 to 99.4 percent. Could the model just be over-confident?

- **Answer:** No, because the confidence rose on correct predictions, while the ablation study shows the confidence on wrong predictions actually went down, for example from 53.25 to 48.02 percent with TTA. This means the model separates right and wrong cases better, not that it is blindly over-confident.
- **Justification:** This shows the difference between better calibration and false confidence.

Slide 13 - Baseline vs Proposed Model

Speaking Script

- This slide quantifies how much the full pipeline improved over a plain baseline.
- The baseline is a single EfficientNetB0 trained with standard settings, with no oversampling, no class weighting, no cosine decay, no SWA, and no TTA.
- That baseline reached 84.91 percent accuracy with 24 errors and a macro F1 of only 79.19 percent.
- My proposed model reached 98.11 percent with just 3 errors and a macro F1 of 97.69 percent.
- That is a 13.2 percentage-point gain in accuracy and an 87.5 percent reduction in errors.
- Crucially, the macro F1 rose almost as much as the accuracy, which proves the improvement is spread fairly across all three classes rather than concentrated in the easy majority classes.
- The gain comes mainly from three things working together: hard-example oversampling, cosine decay scheduling, and the five-seed SWA ensemble.

Expected Panel Questions With Answers

- **Q1:** Why compare with a baseline?
- **Answer:** The baseline shows the starting performance before optimisation.
- **Justification:** This proves the proposed improvements are meaningful.

- **Q2:** Which optimisation helped the most?
- **Answer:** The improvement comes from the full pipeline, especially class imbalance handling and ensemble averaging.
- **Justification:** Each method solves a different weakness.

- **Q3:** What does error reduction mean?
- **Answer:** The model made fewer wrong predictions, from 24 errors to 3 errors.
- **Justification:** This shows better practical reliability.

- **Q4:** Is the comparison between the baseline and the proposed model fair?
- **Answer:** Yes, both use the same dataset, the same stratified split, the same test set, and the same EfficientNetB0 backbone. The only differences are the added steps of class imbalance handling, five-seed SWA, and multi-scale TTA, so the 13.20 percent improvement can be credited to them.

- **Justification:** A fair setup makes the improvement claim valid.

Slide 14 - Ablation Study

Speaking Script

- The ablation study is how I justify my final configuration with evidence rather than assumption.
- I tested four ensemble sizes, one, two, three, and five seeds, each with and without TTA, giving eight configurations from A to H.
- The clearest finding is that ensemble size is the main driver of performance. A single seed gave 6 errors, but three seeds halved that to 3 errors with a macro F1 of 97.69, and there was no further accuracy gain at five seeds.
- So why keep five seeds and TTA? Because once accuracy plateaus, reliability becomes the deciding factor.
- Five seeds average out more of the random initialisation variance, giving a more stable decision boundary, and at five seeds TTA produces the lowest confidence on wrong predictions, about 48 percent, so mistakes are less overconfident.
- Configuration H, five seeds with TTA, was therefore selected as the best balance of accuracy, error count, F1-score, and reliability.

Expected Panel Questions With Answers

- **Q1:** What is the purpose of an ablation study?
- **Answer:** It checks which parts of the method actually improve the result.
- **Justification:** This avoids claiming improvement without evidence.

- **Q2:** Why choose 5-seed plus TTA if accuracy is same as 5-seed?
- **Answer:** Because it reduces wrong prediction confidence, making incorrect answers less overconfident.
- **Justification:** This improves trustworthiness.

- **Q3:** Why does TTA help?
- **Answer:** It tests the same image in multiple views and averages the predictions.
- **Justification:** This makes the final answer less sensitive to one image view.

- **Q4:** The three-seed configuration already reaches 98.11 percent with three errors and the same F1. Why pay the extra cost of five seeds?
- **Answer:** Once accuracy and F1 are saturated, the deciding factor is reliability. Five seeds reduce run-to-run variation, and with TTA they give the lowest wrong-prediction confidence at 48.02 percent, which is the best-calibrated and most

repeatable setting, while the matching three-seed numbers can be partly down to luck on a small test set.

- **Justification:** This defends the choice using stability and calibration, not only accuracy.

Slide 15 - Conclusion & Future Work

Speaking Script

- In conclusion, all of the research objectives were achieved.
- I developed an EfficientNetB0 ensemble that classifies coral health accurately, I added Grad-CAM so every prediction can be explained and validated visually, I evaluated the model rigorously and reached 98.11 percent test accuracy, and I deployed it as a working Flask web application.
- The key achievement is a system that is at once accurate, explainable, and deployable, which previous studies usually treated as separate problems.
- For future work, the most important step is external validation on larger and more diverse reef datasets, especially to grow the small Dead class.
- Beyond that, I would compress the ensemble through distillation or quantisation for edge deployment, add active learning so experts label the most uncertain images, and explore cross-domain adaptation and autonomous underwater or federated approaches for real-world scale.

Expected Panel Questions With Answers

- **Q1:** What is the main contribution of this project?
- **Answer:** The main contribution is an accurate, explainable, and deployable coral health assessment framework.
- **Justification:** It combines prediction, explanation, and deployment.

- **Q2:** What is the most important future work?
- **Answer:** Model compression is important because it can make the ensemble lighter for deployment.
- **Justification:** This helps the system run in real-world environments.

- **Q3:** What is the final takeaway?
- **Answer:** The system can classify coral health accurately and explain the prediction visually.
- **Justification:** This makes it useful and trustworthy for coral assessment.

- **Q4:** If this went towards real deployment, what is the single most important next step?

- **Answer:** The most important step is to validate the model on a larger and more diverse dataset from multiple reefs and to grow the Dead class, so generalisation is confirmed before compressing the ensemble for edge or underwater deployment.
- **Justification:** This prioritises proving generalisation before engineering deployment.

Closing - Summary and Thank You

Speaking Script

- To summarise, this project delivered an EfficientNetB0 coral classifier, added Grad-CAM to explain its decisions, and evaluated it fairly across all three classes.
- The proposed five-seed SWA ensemble with multi-scale TTA reached 98.11 percent test accuracy with only three errors, a clear improvement over the baseline.
- All three research objectives were therefore achieved: the system is accurate, explainable, and deployable.
- Thank you for your time and attention. I am happy to answer any questions from the panel.

Expected Panel Questions With Answers

- **Q1:** What is the single biggest limitation of your work?
- **Answer:** The main limitation is the small and imbalanced Dead class, especially the 15 test images, which makes the Dead metrics less certain than the others.
- **Justification:** Naming the key limitation clearly shows mature self-evaluation.

- **Q2:** What is your main contribution in one sentence?
- **Answer:** The main contribution is a single coral health framework that combines high accuracy, visual explainability, and practical deployment, which previous studies usually addressed separately.
- **Justification:** This restates the research gap that the project fills.

Appendix A - 15-Minute Presentation Script (Strict Timing)

Tight speaking notes - aim to finish each slide by the time shown

- **Opening (by 0:30):** Good morning, panel. My project is Coral Health AI, a deep learning system that classifies coral as Healthy, Bleached, or Dead, explains each decision with Grad-CAM, and reaches 98.11 percent accuracy. I will cover the problem, the method, the results, and the conclusion.
- **Slide 1 - Problem (by 1:30):** Coral monitoring today has three problems: manual assessment is slow and expert-dependent; CNN models are black boxes users cannot trust; and accuracy alone hides weak performance on small classes. I solve all three with an accurate, explainable, and well-evaluated framework.
- **Slide 2 - Objectives (by 2:15):** Three objectives: build an EfficientNetB0 classifier, add Grad-CAM for visual explanation, and evaluate with precision, recall, and F1, making the model accurate, explainable, and reliable.
- **Slide 3 - Scope (by 3:00):** The scope is image-based classification of the BHD dataset into three classes, covering preprocessing, training, evaluation, Grad-CAM, and Flask deployment. It does not use water-quality or sensor data.
- **Slide 4 - Literature & Gap (by 3:45):** The literature supports EfficientNetB0, ensembling, multi-scale TTA, and Grad-CAM individually. The gap is that no prior coral study combines all four. That combination is my contribution.
- **Slide 5 - Literature Summary (by 4:15):** These recent 2021-to-2025 studies each back one design choice, EfficientNet for accuracy, ensembles for robustness, TTA for stable inference, and Grad-CAM for explainability, so my design is evidence based.
- **Slide 6 - Methodology (by 5:30):** Three phases. Data preparation: resize, stratified split, augment. Model development: train EfficientNetB0, five-seed SWA, multi-scale TTA. Analysis and deployment: evaluate, Grad-CAM, deploy in Flask. Augmentation is applied only to training, so there is no data leakage.
- **Slide 7 - Dataset (by 6:45):** 1,582 images: 712 Healthy, 720 Bleached, but only 150 Dead, so it is imbalanced. Images are resized to 224. I handle imbalance with augmentation, hard-example oversampling, and a mild 1.3 class weight on Dead.
- **Slide 8 - Model (by 8:15):** EfficientNetB0 with ImageNet transfer learning and dropout. SWA averages the weights from epochs 26 to 30 into one stable model; five seeds are ensembled; and at inference, multi-scale TTA averages four views, the 224 and 256 sizes each original and horizontally flipped, across the five models for stable predictions.
- **Slide 9 - Evaluation (by 9:00):** I test on 159 held-out images never seen in training, and report accuracy, macro precision, recall, F1, and the confusion matrix, so performance is judged both overall and per class.

- **Slide 10 - Grad-CAM (by 9:45):** Grad-CAM turns the final convolutional layer into a heatmap, where red is high focus. For healthy coral it should land on texture and colour; for bleached and dead, on pale or exposed regions.
- **Slide 11 - Results (by 11:15):** 98.11 percent accuracy, 156 of 159 correct, only 3 errors, and macro F1 of 97.69. Healthy had perfect recall. The three errors are near-boundary: two Bleached read as Healthy, one Dead as Bleached. Macro F1 matters most because it is fair to the small Dead class.
- **Slide 12 - Grad-CAM Results (by 12:00):** Versus the baseline, the proposed ensemble produces tighter heatmaps on coral and far higher confidence, for example bleached confidence rises from 45.5 to 99.4 percent. So it is more accurate and more explainable.
- **Slide 13 - Baseline vs Proposed (by 12:45):** Same data, same split, same backbone. The baseline was 84.91 percent with 24 errors; the proposed model is 98.11 with 3, a 13.2-point gain and 87.5 percent fewer errors, from imbalance handling, SWA, and TTA.
- **Slide 14 - Ablation (by 13:30):** I tested one to five seeds, with and without TTA. Five-seed plus TTA gives the best balance: the same top accuracy but the lowest wrong-prediction confidence, so it is the most reliable. That is why I selected configuration H.
- **Slide 15 - Conclusion (by 14:15):** All three objectives are met: an accurate EfficientNetB0 classifier, Grad-CAM explanations, and rigorous evaluation at 98.11 percent. Future work is model compression, active learning, cross-domain adaptation, and AUV or federated deployment.
- **Closing (by 14:45):** In short, the system classifies coral health accurately and explains why, making it useful and trustworthy. Thank you; I am happy to take questions.
- **Pacing note:** This runs to about 14:45, leaving a small buffer. If you are running late, shorten Slides 4, 5, 10, and 12 first and protect Slides 11 and 14.

Appendix B - Model Answers for the Toughest 5 Questions

Polished, examiner-ready phrasing; 'If pushed' handles the follow-up

- **Question:** Q1 - Dead class has only 15 test images. How reliable is the Dead recall? (Slide 7)
- **Model answer:** That is a fair challenge, and I want to be honest about it. With only 15 Dead test images, the 93.3 percent recall means 14 of 15 correct, so a single error moves the figure by about 6.7 percent; the metric genuinely carries wide uncertainty. That is exactly why I do not rely on accuracy alone: I report macro F1 and the full confusion matrix so the small class cannot be hidden, and in training I used hard-example oversampling and class weighting to protect it. The honest conclusion is that the Dead result is promising but needs more data to confirm, which is why collecting more Dead images is my first future-work item.
- **If pushed:** The 98.11 percent overall figure is reliable because Healthy and Bleached each have 72 test images; it is specifically the Dead per-class number that has wider error bars, and I have stated that openly rather than overclaiming.

- **Question:** Q2 - Two Bleached corals were predicted as Healthy. Is that not the most costly error? (Slide 11)
- **Model answer:** You have identified the most important cell in the whole matrix. Two Bleached corals were predicted as Healthy, and operationally that is the worst kind of mistake, because a stressed reef flagged as fine could be left without intervention. I address it in three ways. First, I report per-class recall, not just accuracy, so this error is visible. Second, the system always shows the Grad-CAM heatmap and the confidence score, so a human can check a Healthy call before acting. Third, in deployment I would lower the decision threshold for Healthy or route low-confidence cases to expert review, because in conservation a false Healthy is costlier than a false Bleached.
- **If pushed:** Yes, I would deliberately make the model err toward caution: flagging a healthy reef for a second look is cheap, but missing a bleaching event is not.

- **Question:** Q3 - Explain precisely the difference between SWA, the five-seed ensemble, and TTA. (Slide 8)
- **Model answer:** They operate at three different stages. SWA, Stochastic Weight Averaging, works on a single model: instead of taking the final epoch, I average the weights from epochs 26 to 30 into one model that sits in a flatter, more stable minimum. The ensemble works across models: I train five such SWA models from five different random seeds and average their output probabilities, which cancels the randomness of any single run. TTA, Test-Time Augmentation, works at inference on

the input: each image is run at two sizes, 224 and 256 with the 256 centre-cropped back to 224, each original and horizontally flipped, giving four views whose predictions I average across the five models, twenty predictions in total. So SWA averages weights, the ensemble averages models, and TTA averages views.

- **If pushed:** They are three kinds of averaging, but over three independent sources of variance, the training trajectory, the random seed, and the input view, and averaging each is what drives the error count from 24 down to 3.
- **Question:** Q4 - Three seeds already reach 98.11 percent. Why pay for five seeds? (Slide 14)
- **Model answer:** You are right that three seeds already reach 98.11 percent with three errors and the same macro F1, so accuracy has saturated. The deciding factor is therefore not accuracy but reliability. On a test set this small, identical headline numbers can partly be seed luck, so I want the configuration that is least sensitive to randomness. Five seeds reduce run-to-run variance more than three, and with TTA the five-seed setting also gives the lowest confidence on its wrong predictions, about 48 percent, meaning that when it is wrong it is less dangerously over-confident. I chose configuration H for that calibration and stability.
- **If pushed:** It is only a training-time cost, done once and offline; at inference the ensemble runs as a single averaged prediction, so deployment is not affected.
- **Question:** Q5 - 98 percent on a small test set looks too good. How do you rule out overfitting? (Slide 11)
- **Model answer:** It is a healthy thing to question, so let me give the evidence rather than just assert it. First, there is no data leakage: the split is stratified and done at the image level before any augmentation, and the 159 test images are never seen in training or validation. Second, I use several anti-overfitting measures, ImageNet transfer learning, dropout, augmentation, plus SWA and ensembling, which reduce variance rather than memorise. Third, the ablation study shows the gains are consistent across configurations, not a single lucky run. The one caveat I repeat is the small Dead test set, which I treat with more caution than the overall number.
- **If pushed:** No, I did not tune on the test set: all tuning, including the SWA epoch range and the class weights, was done on a separate validation set, and the test set was used only once, at the end.

Appendix C - Unexpected & Report-Bridge Questions

Lateral 'curveball' questions panels ask - the easy ones to miss

- Your full report-based bank, about 70 questions across architecture, training, TTA, dataset, Grad-CAM, results, ablation, deployment, and methodology, is in Viva_QA_Preparation.docx under the Deliverables folder. The items below are the gaps and curveballs that the slides and that bank do not emphasise.
- **Q1:** What happens if I upload a non-coral image, for example a fish, a person, or a blank photo, to your web app?
- **Answer:** The model uses a three-class softmax, so it will always force the image into Healthy, Bleached, or Dead, usually with low or unstable confidence. It has no 'not coral' or 'unknown' option.
- **Justification:** Honest out-of-distribution behaviour; the mitigation is a confidence threshold or a reject / unknown class as future work.
- **Q2:** Who labelled the BHD dataset, and how do you know the ground-truth labels are correct?
- **Answer:** The labels come with the public Kaggle BHD Coral Dataset by Jamil 2021; I did not independently re-verify them with a marine biologist, so the model can only be as accurate as those labels.
- **Justification:** Acknowledging label provenance and its limit is mature; expert re-labelling is a valid improvement.
- **Q3:** Coral bleaching is a continuous process, not three fixed states. How does a three-class model handle a partially bleached coral?
- **Answer:** The model assigns the single most probable class and exposes the confidence, so a borderline coral usually shows a low, split confidence rather than a confident wrong answer. Capturing true severity would need an ordinal or regression model.
- **Justification:** Shows I understand the biology is a continuum and where the model stops.
- **Q4:** If I gave you a coral photo from a different ocean and a different camera right now, would it still work?
- **Answer:** I cannot guarantee it, because the model was trained and tested only on the BHD dataset from limited regions, so performance under domain shift is unverified. That is exactly why external validation is my main requirement before deployment.

- **Justification:** Directly and honestly addresses generalisation, matching the discussion chapter.
- **Q5:** Your headline is 98.11 percent. What is the confidence interval on 159 samples?
- **Answer:** On 159 samples a 95 percent Wilson interval is roughly plus or minus two percent, so about 95 to 99.5 percent. For the Dead class on only 15 samples the interval is far wider, which is why I treat the Dead figure with caution.
- **Justification:** Gives an actual error bar instead of a bare point estimate.
- **Q6:** Isn't a five-seed ensemble just hiding that a single model is not good enough?
- **Answer:** A single model already reaches 84.91 percent; the ensemble is not a crutch but a variance-reduction method that averages five independently initialised SWA models to smooth the decision boundary near ambiguous cases. The ablation shows the gain is real and saturates by three to five seeds.
- **Justification:** Reframes ensembling as a principled, evidence-backed choice.
- **Q7:** What did you actually build, versus what came from pretrained models and libraries?
- **Answer:** The EfficientNetB0 backbone and ImageNet weights are pretrained, and Keras and TensorFlow provide the building blocks. My contribution is the full pipeline: the custom head, hard-example oversampling at 30 times, five-seed SWA, multi-scale TTA, Grad-CAM integration, the ablation that justifies the configuration, and the Flask deployment.
- **Justification:** Cleanly separates reuse from original contribution, which panels probe.
- **Q8:** What happens after deployment if the model quietly gets worse over time, that is model drift?
- **Answer:** My current system has no drift monitoring; in operation I would log predictions and confidence, periodically have experts review low-confidence and random samples, and retrain when accuracy on that reviewed data drops.
- **Justification:** Shows awareness of the maintenance and monitoring dimension, not just training.
- **Q9:** If you had to delete one component from your pipeline, which one and why?

- **Answer:** I would drop multi-scale TTA first, because the ablation shows it is accuracy-neutral at five seeds and I keep it only for robustness. I would never drop the five-seed SWA ensemble or the oversampling, which produce most of the 13.2-point gain.
- **Justification:** Demonstrates I know each component's real contribution.
- **Q10:** Your inference is 10.38 milliseconds on an RTX 3070. What about on a boat with a phone or a Raspberry Pi?
- **Answer:** That figure is for five models across four TTA views on a desktop GPU, so on edge hardware it would be far slower. For field use I would compress the ensemble through knowledge distillation or quantisation into one lightweight model, which is in my future work.
- **Justification:** Connects the measured speed to realistic deployment constraints.
- **Q11:** Could your model be fooled, for example by an adversarial or heavily edited image?
- **Answer:** I did not test adversarial robustness, so I cannot claim it is secure; like most CNNs it could probably be fooled by crafted perturbations. For an ecological monitoring tool the realistic risk is low, but it is an untested limitation.
- **Justification:** Honestly scopes robustness rather than overclaiming.
- **Q12:** What was the single hardest problem in this project and how did you solve it?
- **Answer:** The hardest problem was the tiny, imbalanced Dead class. I solved it with hard-example oversampling at 30 times, a 1.3 times class weight, and macro-averaged evaluation, which lifted Dead precision to 1.000 and recall to 93.3 percent, though I accept more Dead data is still needed.
- **Justification:** A specific, genuine reflection scores well on problem solving.
- **Q13:** Why image classification instead of segmentation or object detection?
- **Answer:** My research question is the overall health condition of a coral image, which is a whole-image classification task. Segmentation would need pixel-level masks the dataset does not provide, and detection would need bounding boxes; patch-level or segmentation analysis is a natural extension.
- **Justification:** Justifies the task framing against the obvious alternatives.
- **Q14:** Would you personally trust this to decide real conservation spending?

- **Answer:** I would trust it as a fast first-pass screening tool that flags reefs and shows its Grad-CAM reasoning, but not as the sole decision-maker. Final conservation decisions should keep a human expert in the loop, especially for low-confidence or Bleached-to-Healthy cases.
- **Justification:** Positions the system as decision support, which is the defensible claim.

Slide-vs-Report Bridge (resolve it before the panel does)

- **Q15:** Your report lists four objectives, but your slides show three. Why the difference?
- **Answer:** The three on the objectives slide are the core modelling objectives: build the EfficientNetB0 model, add Grad-CAM, and evaluate it. The fourth objective in the report is deployment, which I demonstrate through the Flask web application shown in the scope and conclusion slides. All four objectives are achieved.
- **Justification:** Resolves a real slide-versus-report mismatch before the panel raises it.
- Report-only facts to have ready (not on the slides): random seeds 42 to 46; about 12 minutes per seed and about 60 minutes total on an RTX 3070 8 GB (CUDA 11.8, cuDNN 8.6); 10.38 milliseconds per image at inference; dropout 0.4; L2 weight decay lambda 0.0002; label smoothing 0.05; Adam optimiser at learning rate 8×10^{-5} with cosine decay; batch size 16; 30 epochs; final 100 backbone layers fine-tuned; Grad-CAM taken from the top_conv layer; TensorFlow 2.12 on Python 3.10.